

Mitsue Village — Forest Area: Before / After (25-Year Projection)

Prepared for: 上坂奈穂様 (Nao Uesaka), Mizuho Securities — Sustainability Promotion Dept. **Purpose:** Order-of-magnitude estimate of how much sugi/hinoki plantation area the project's biomass-CHP-funded workforce could realistically thin and convert toward native broadleaf over 25 years, and what that implies for energy output.

Status: These are modeled projections from the project's Forest Twin calculation, calibrated against a literature benchmark (Mishima, Fukushima — NIES 2020, ~750 t/yr feeding ≤50 kWe). **Correction (2026-09-07):** NIES confirmed by email that the Mishima paper is a simulation, not an installed system — this benchmark was not previously verified as "empirical" when this document was sent; it is a modeled scenario figure, not measured data. They are not measured or committed figures.

Before (today)

Item	Value
Total village forest	~7,051 ha (>90% privately owned, postwar sugi/hinoki plantation)
Annual thinning (current co-op capacity)	~66 ha/yr
Wood actually recovered	only ~120–300 m³/yr
Forest cooperative crew	small, average age ~33

The gap: thinning today is subsidy-driven and leaves the cut wood largely in the forest — it is not a mechanized harvest, so almost none of the cleared area is followed by active broadleaf replanting.

The mechanism

Biomass-CHP revenue from harvested sugi thinnings pays for a larger, mechanized cooperative crew (harvester + forwarder, chipper, drying). A bigger crew thins more forest faster; the cleared area is where native broadleaf replanting (see companion species list) becomes viable, restoring wildlife forage and reducing deer/boar crop damage. The same harvested wood fuels the CHP plant that powers the AI data center — energy output and forest restoration are the same lever, not two separate programs.

After — 25-year scenarios by workforce scale

Cooperative capacity	Harvest rate	Area converted in 25 yr	Wood/yr	Biomass plant	Electricity/yr
Current + light mechanisation	~70 ha/yr	1,762 ha (25% of forest)	10,500 m³	~290 kWe	~2,000 MWh
Double crew + harvester/forwarder — recommended baseline	~140 ha/yr	3,526 ha (50% of forest)	42,000 m³	~1,150 kWe	~8,000 MWh

Cooperative capacity	Harvest rate	Area converted in 25 yr	Wood/yr	Biomass plant	Electricity/yr
Triple crew, full mechanisation	~200 ha/yr	~5,000 ha (~70%)	~60,000 m³	~1.6 MWe	~11,000 MWh
Optimistic (~4× crew)	254 ha/yr	6,346 ha (90%)	135,000 m³	3.7 MWe	~26,000 MWh

Recommended baseline: doubled workforce — roughly half the village's forest (~3,500 ha) thinned and opened to broadleaf restoration over 25 years, supporting a ~1.1–1.2 MWe biomass plant.

Precedents used to sanity-check this

Village	Forest	Harvest model	Result
Shimokawa, Hokkaido (下川町)	4,500+ ha	~50 ha/yr, 60-yr sustained cycle	56% local heat self-sufficiency (real, operating)
Nishiawakura, Okayama (西粟倉村)	93–95% forest	~3,000 ha under 100-yr management	forestry-jobs revival (real, operating)
Mishima, Fukushima (三島町)	~88% forest	~750 t/yr (simulated scenario)	≤50 kWe — our model's calibration anchor; NIES-confirmed simulation, not a built system

Sources

- Mitsue Forest Workforce & Energy Plan (project document, v1.7, 2026-07-17)
- 地域林業パートナー cooperative report
- NIES 2020 — Mishima, Fukushima biomass CHP simulation study (confirmed by NIES email, 2026-09-07, as a modeled scenario, not an installed system)
- Shimokawa Town / Nishiawakura Village public forestry data